

Efficiency (%)

97
96
95
94
93
92
91
90
89
88

IR
Sup/IRBuck

Exceed 96% Peak Efficiency With Gen2.1 Sup/IRBuck™ Integrated Voltage Regulators

IR's Gen2.1 Sup/IRBuck™ devices save time, space and energy for your POL design



IR384XW/3XW Sup/IRBuck™ FAMILY IMPROVEMENTS

- Added Over Voltage Detection (PGOOD window comparator)

- Shorter dead-time to reduce power loss
- Reduced SYNC FET $R_{DS(on)}$ maximum value for better over-current limit accuracy
- New current options

FEATURES:

- Wide input voltage range (1.5V to 16V with 5V bias)
- Small footprint (5x6mm) and low height (0.9mm)
- Programmable frequency up to 1.5MHz
- 1% accurate 0.7V reference voltage
- Programmable hiccup current limit and soft start
- Enhanced pre-bias start up
- Thermal protection
- Enable pin with voltage monitoring capability
- Power Good output for over-voltage and under-voltage detection
- Optimized solutions for sequencing (IR3840/1/2/3W) and DDR memory tracking (IR3831/32W)
- -40°C to 125°C operating junction temperature (Tj)
- Pin compatible with Gen2 Sup/IRBuck products

Part Number	Package	Input Voltage	Maximum Current	Maximum Frequency	Additional Features
IR3843WM	5x6 PQFN	1.5V to 16V	2A	1.5MHz	OV detection, Sequencing
IR3842WM	5x6 PQFN	1.5V to 16V	4A	1.5MHz	OV detection, Sequencing
IR3841WM	5x6 PQFN	1.5V to 16V	8A	1.5MHz	OV detection, Sequencing
IR3840WM	5x6 PQFN	1.5V to 16V	12A	1.5MHz	OV detection, Sequencing
IR3832WM	5x6 PQFN	1.5V to 16V	4A	1.5MHz	OV detection, DDR tracking
IR3831WM	5x6 PQFN	1.5V to 16V	8A	1.5MHz	OV detection, DDR tracking

For more information please visit us at www.irf.com or www.irf.com.cn

Sup/IRBuck™ is a trademark of International Rectifier

International
IR Rectifier
THE POWER MANAGEMENT LEADER